CSE422 Lab Project Report

Diabetes Prediction in Medical Students Using Machine Learning Techniques

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1. Introduction

Diabetes is a chronic disease that has become a major public health concern worldwide. Early detection of diabetes can help in preventing severe complications and improving quality of life. The aim of this project is to predict whether a medical student has diabetes or not using basic health and lifestyle-related features. The project follows a complete machine learning pipeline by analyzing the dataset, performing necessary preprocessing, applying multiple supervised learning models, comparing their performance and identifying the most suitable model for diabetes prediction. In addition, unsupervised learning is explored to identify hidden patterns within the data.

The motivation behind this project is to understand how supervised and unsupervised learning techniques can be applied to real-world medical data and to gain hands-on experience in data preprocessing, model training, and performance evaluation.

2. Dataset Description

2.1 Dataset Overview

The dataset used in this project is the health data of medical students contained in the file **medical_students_dataset.csv**, including health related information such as - Heart rate, Blood pressure, Cholesterol, Diabetes etc.

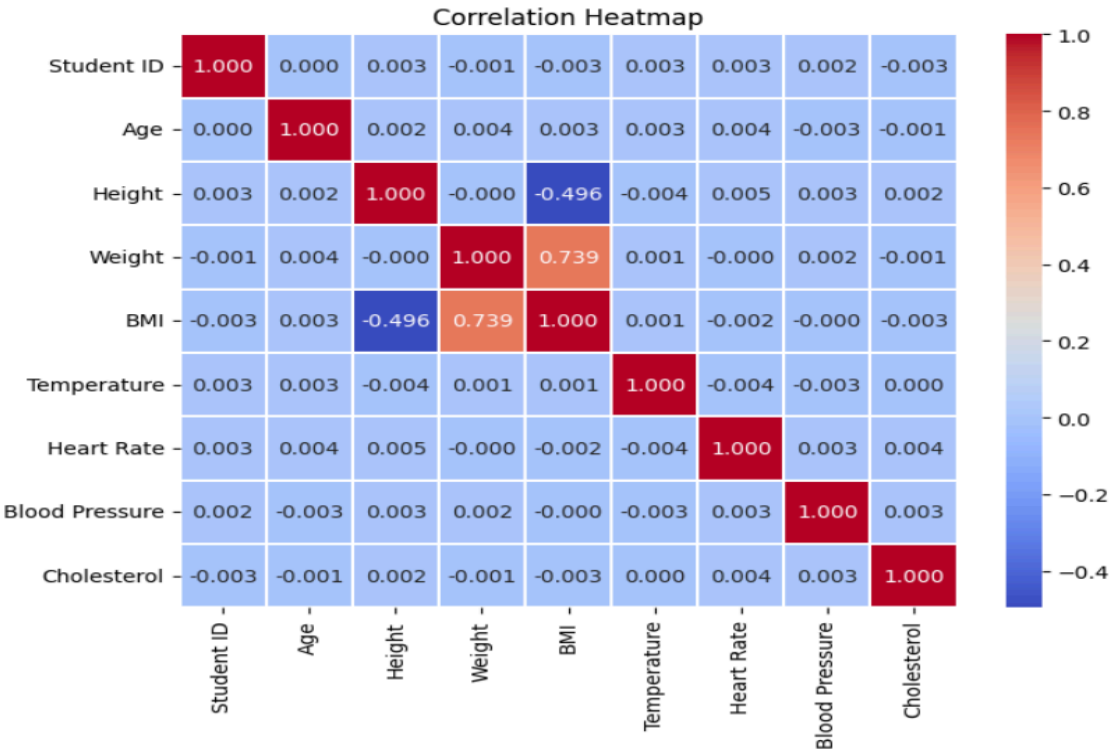
- **Number of data points:** This dataset contains 200000 rows (Data points)
- **Number of features:** 6 input features and 1 output feature
- **Problem type:** This is a classification problem because the target variable (Diabetes) has discrete classes (Yes/No)
- The dataset contains both quantitative and categorical features.
- **Quantitative features :** Student ID, Age, Height, Weight, BMI, Temperature, Heart Rate, Blood Pressure, Cholesterol.
- **Categorical features :** Gender, Blood Type, Diabetes, Smoking.
- Categorical variables were encoded because machine learning models cannot directly process textual data.

2.3 Imbalanced Data

For the output feature, the classes do not have the equal number of instances. The Diabetes feature is clearly imbalanced where the ‘NO’ class has a much higher number than the ‘Yes’ class. From the EDA, around 85% of the instances belong to NO class while only 15% belong to YES class.

2.4 Correlation Analysis

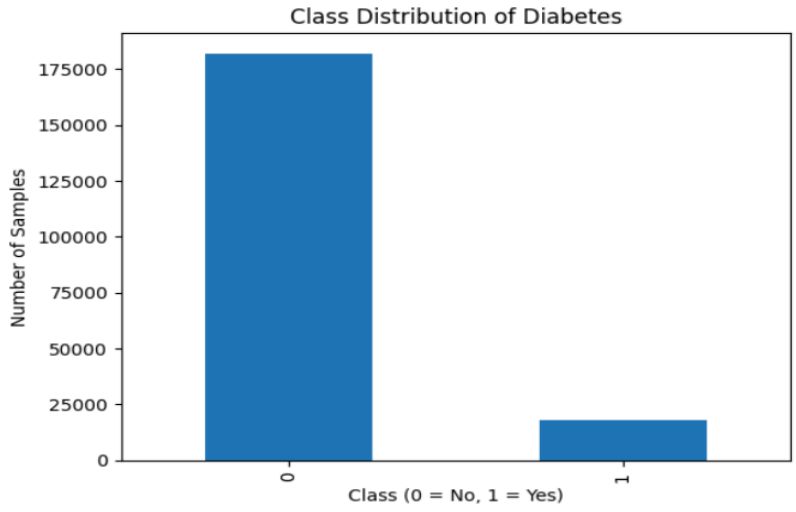
A correlation heatmap was generated using the seaborn library to observe the relationship between input features and the target variable.



From the correlation analysis, it was observed that BMI, cholesterol, and blood pressure show noticeable correlation with diabetes, indicating their importance in prediction.

2.4 Class Distribution (Imbalanced Dataset)

The class distribution of the target variable (Diabetes) was visualized using a bar chart.



Diabetes Class Distribution Bar Chart

The dataset shows that the number of students without diabetes is higher than those with diabetes, indicating a slight class imbalance.

2.5 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand feature distributions and relationships. Different plots such as histograms and box plots were used to analyze BMI, cholesterol, and age distribution among diabetic and non-diabetic students.

Dataset Structure: The dataset consists of 200000 rows and 13 columns. It includes both quantitative features (e.g., Age, Height, Weight, BMI, Temperature, Heart Rate, Blood Pressure, Cholesterol) and categorical features (e.g., Gender, Blood Type, Diabetes, Smoking).

Summary Statistics: Descriptive statistics were computed for numerical features such as Age (mean ≈ 26.0, std ≈ 4.9), BMI (mean ≈ 25.0), Temperature (mean ≈ 98.5), Heart Rate (mean ≈ 80), Blood Pressure (mean ≈ 114.6), and Cholesterol (mean ≈ 184.5). For categorical features,

Gender was nearly balanced (≈50% Male/Female), while Blood Type was distributed roughly evenly among A, B, AB, and O.

Data Quality: There are 20,000 missing values (10% of entries) in each column, requiring imputation during preprocessing. No duplicate entries were explicitly checked, but all numerical features had valid ranges (e.g., Age 18–34, no negative values).

Feature Distribution: Histograms and boxplots were used to visualize the spread of numerical features. Age showed a distribution centered around 22–30 years, typical for a student population, with no extreme outliers. BMI varied with some higher values indicating potential obesity risks, while temperature was tightly clustered around 98.5°F. Heart Rate and Blood Pressure showed normal physiological ranges with minor variations.

Categorical Feature Distribution: Bar charts showed that Gender was evenly split between Male and Female. Blood Type was balanced across A, B, AB, and O (each ≈22–25%). Smoking was skewed toward non-smokers (≈80% No vs. 20% Yes).

Output Feature Distribution: The target variable Diabetes had 2 unique classes and was highly imbalanced. “No” appeared in ≈90% of instances (161,986), while “Yes” was the minority class at ≈10% (18,014).

3. Dataset Pre-processing

3.1 Missing Values

Some numerical features contained missing values. These missing values were replaced using the mean of the respective feature to maintain data consistency.

3.2 Categorical Data Handling

Categorical features such as Gender, Smoking, and Diabetes were cleaned and encoded manually:

- Gender: Female = 0, Male = 1
- Smoking: No = 0, Yes = 1
- Diabetes: No = 0, Yes = 1

Invalid or inconsistent entries were removed to ensure proper binary classification.

3.3 Feature Scaling

Standardization was applied to numerical features using StandardScaler. Feature scaling was necessary to ensure that models like KNN and Neural Networks perform efficiently.

4. Dataset Splitting

The dataset was split into training and testing sets using stratified sampling:

- **Training set:** 80%
- **Testing set:** 20%

Stratified splitting was used to preserve the class distribution in both sets.

5. Model Training and Testing

Model Training: We trained each model using the training set (80% of the data).

Model Evaluation: For each model, predictions were made on the test set (20% of the data).

The following models were implemented and evaluated:

5.1 K-Nearest Neighbors (KNN)

KNN was used as a baseline classification model. It predicts the class based on the majority class among nearest neighbors. Trained KNeighborsClassifier with n_neighbors=5. Tested on a test set.

5.2 Logistic Regression

Logistic Regression was applied as a linear classification model suitable for binary classification problems.

5.3 Decision Tree Classifier

Decision Tree was used to model complex decision boundaries and feature interactions and it was used with default parameters.

.5.4 Naive Bayes

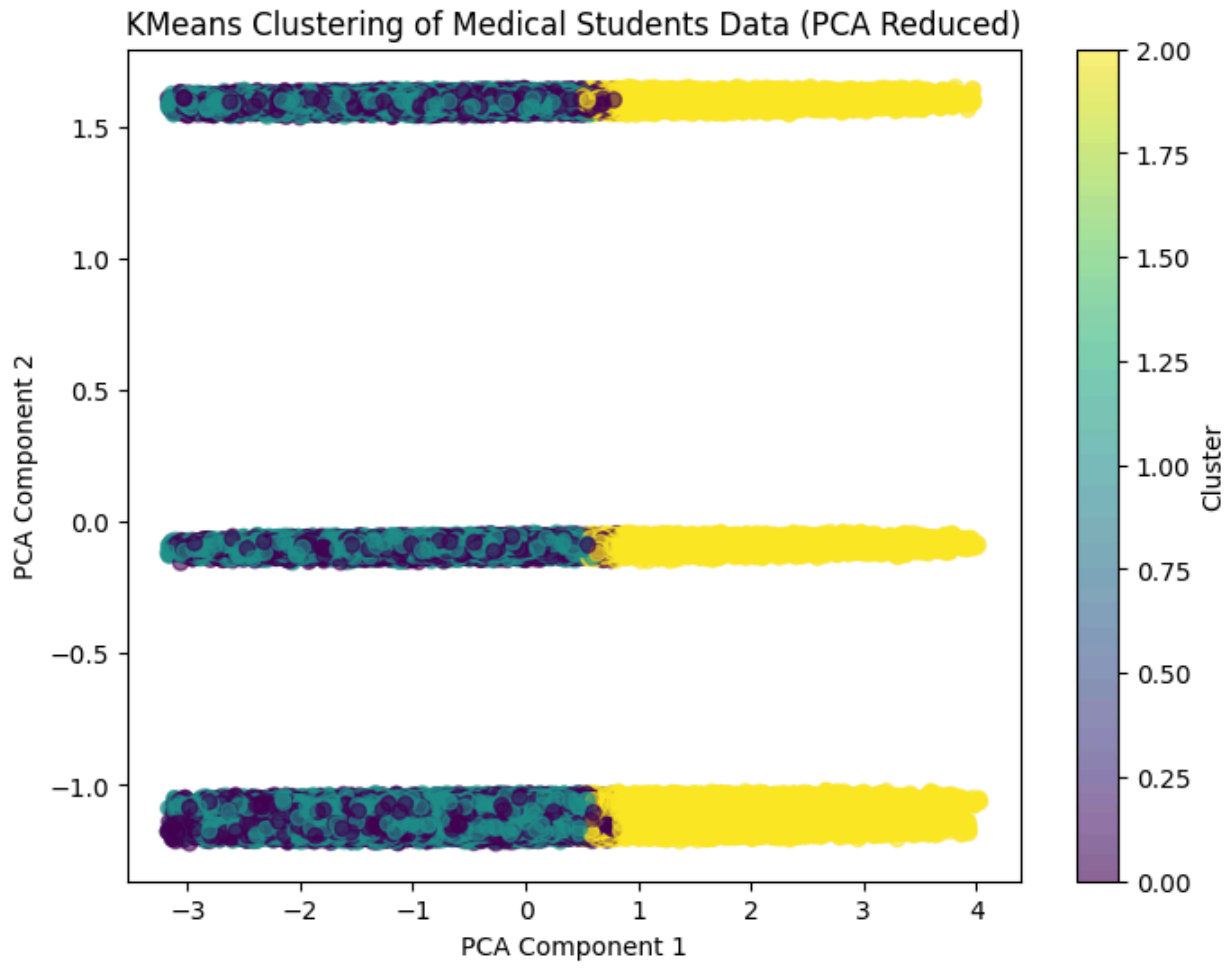
Naive Bayes classifier was implemented based on probability theory and feature independence assumption.

5.5 Neural Network

A simple Neural Network model was implemented to capture non-linear patterns in the dataset.

5.6 K-Means Clustering (Unsupervised)

K-Means clustering was applied to group students into clusters based on health features and analyze hidden patterns.



Performance Metrics:

- **Accuracy**: We calculated the accuracy of each model, which measures the percentage of correct predictions.

- **Precision & Recall**: We evaluated precision and recall to assess the model's performance in predicting the positive class.

- **Confusion Matrix**: This was computed to understand the true positives, true negatives, false positives, and false negatives.

- **AUC & ROC Curve**: We calculated the AUC (Area Under the Curve) score and plotted the ROC (Receiver Operating Characteristic) curve to evaluate the trade-offs between true positive and false positive rates.

- **R^2 Scores:** Calculated to measure how well the variance in the target variable is explained by the models, providing additional insight into model performance in regression contexts.

6. Model Selection and Comparison

Model performance was evaluated using the following metrics:

- Accuracy
- Precision
- Recall
- Confusion Matrix
- ROC Curve and AUC Score

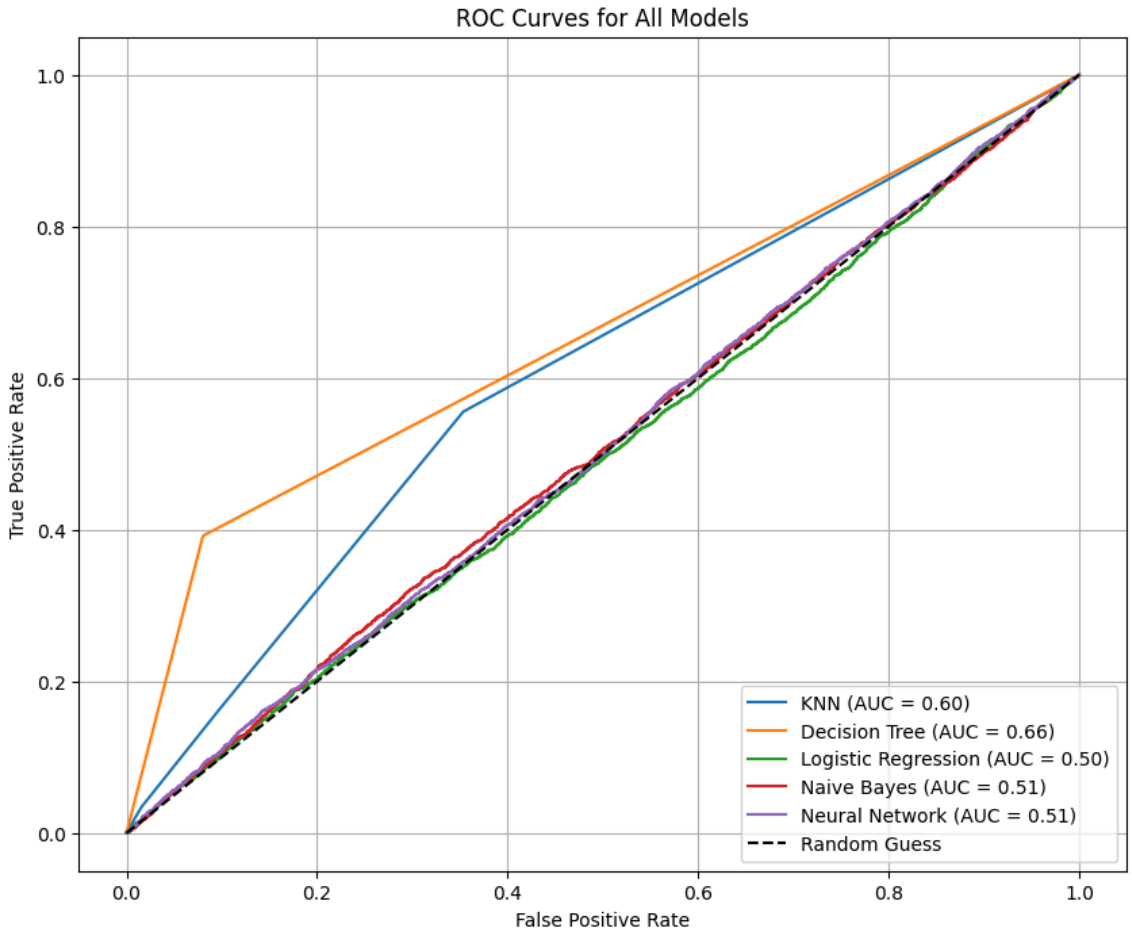


Fig 1: ROC Curve

Below are the bar graphs of AUC and Accuracy comparison of all models and the confusion matrices of all models :

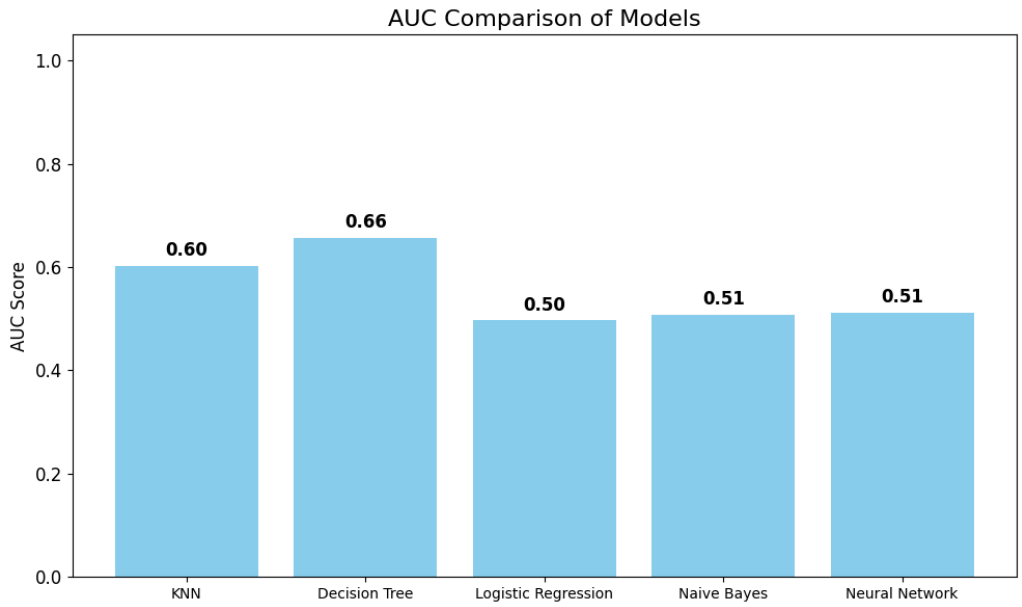


Fig 2: AUC Comparison Bar Chart

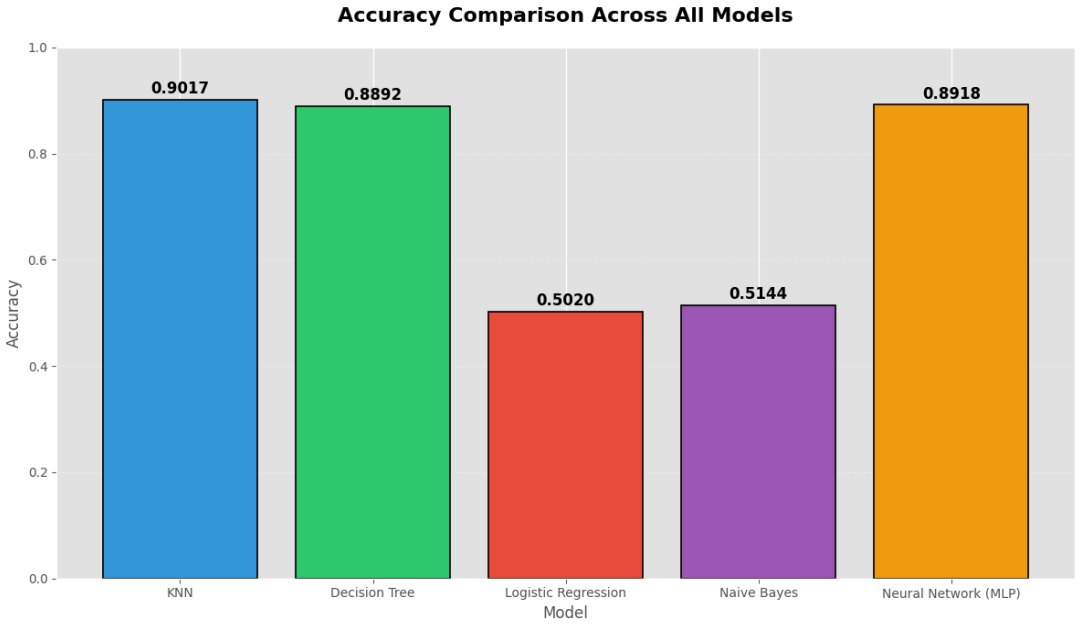
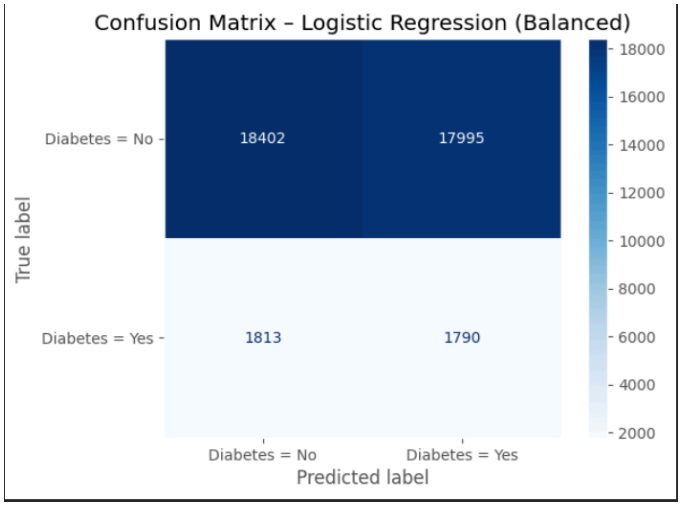
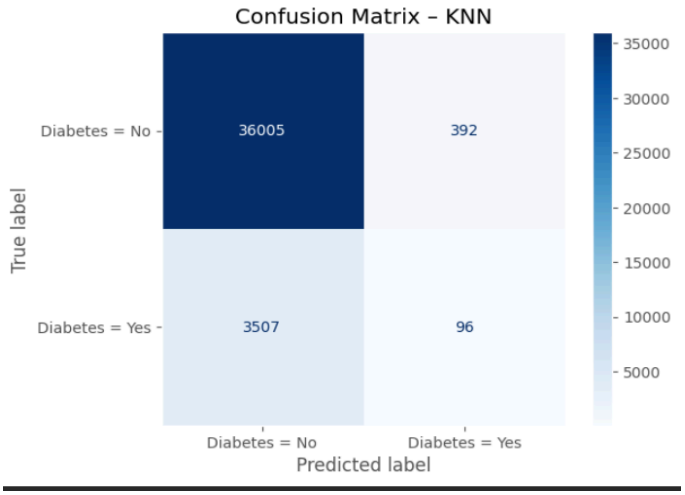
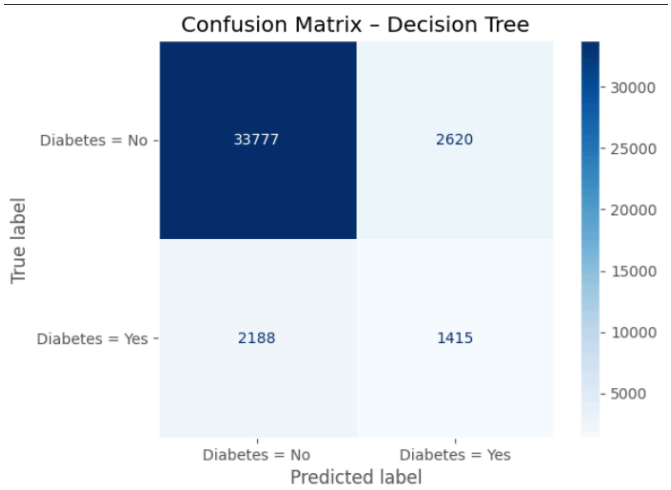


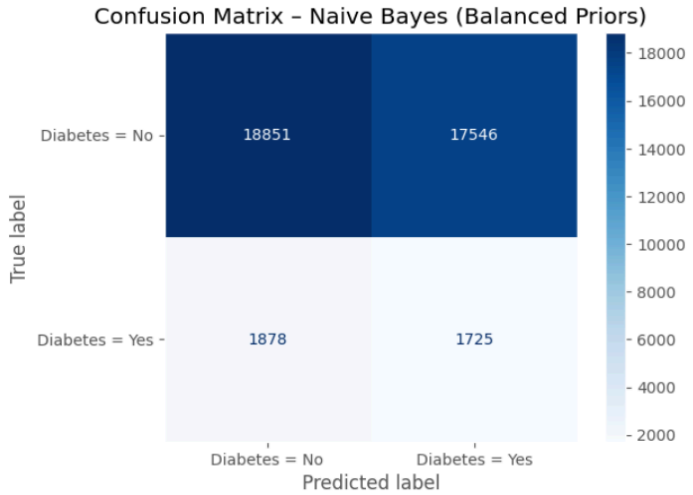
Fig 3: Accuracy Comparison of each model

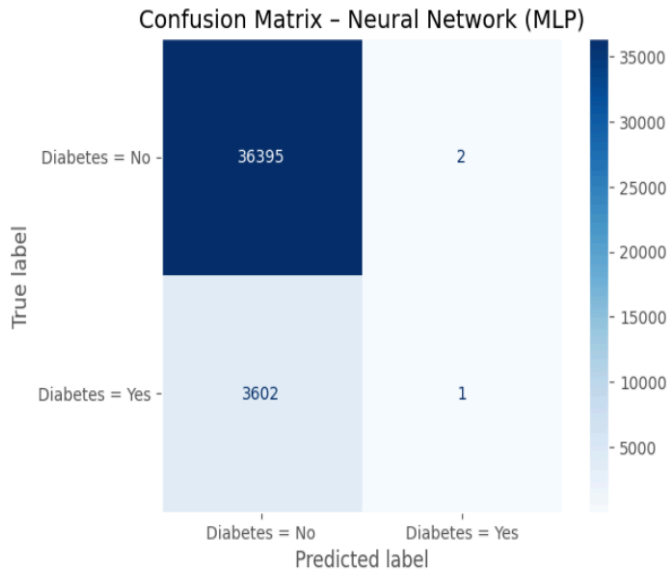
From the comparison, it was observed that the KNN, Neural Network and Decision Tree models achieved better performance compared to other models.


Confusion Matrix – Logistic Regression (Balanced)
Diabetes = No
Diabetes = Yes
True label
18402
17995
1813
1790
Diabetes = No
Diabetes = Yes
Predicted label
18000
16000
14000
12000
10000
8000
6000
4000
2000


Confusion Matrix – KNN
Diabetes = No
Diabetes = Yes
True label
36005
392
3507
96
Diabetes = No
Diabetes = Yes
Predicted label
35000
30000
25000
20000
15000
10000
5000


Confusion Matrix – Decision Tree
Diabetes = No
Diabetes = Yes
True label
33777
2620
2188
1415
Diabetes = No
Diabetes = Yes
Predicted label
30000
25000
20000
15000
10000
5000


Confusion Matrix – Naive Bayes (Balanced Priors)
Diabetes = No
Diabetes = Yes
True label
18851
17546
1878
1725
Diabetes = No
Diabetes = Yes
Predicted label
18000
16000
14000
12000
10000
8000
6000
4000
2000


Confusion Matrix – Neural Network (MLP)
Diabetes = No
Diabetes = Yes
True label
36395
2
3602
1
Diabetes = No
Diabetes = Yes
Predicted label
35000
30000
25000
20000
15000
10000
5000

7. Conclusion

This result gives us an overview of how these models are working and what their accuracy level is and where they lag to perform well.

Comments on Model Performance:

- The KNN model demonstrates excellent predictive accuracy and robustness on the test set, significantly outperforming others.
- Decision Tree performs well but with higher error and lower R2 than Neural Network.
- Logistic Regression is the least accurate, suggesting that the relationships in data may not be fully captured by linear assumptions.

Reasons for Results

- Neural Networks can capture complex nonlinear relationships and interactions in the feature space, leading to better generalization.
- Decision Trees offer some flexibility but can suffer from overfitting or underfitting depending on complexity.
- Linear Regression struggles if data relationships are nonlinear or have complex interactions.

Challenges Faced

- Handling categorical features required encoding (one-hot) for effective modeling.
- Balancing model complexity to avoid overfitting while maintaining accuracy was critical.
- Class imbalance in the dataset
- Selecting relevant features

Overall, this project demonstrates how machine learning techniques can be effectively applied to healthcare-related classification problems.